Department of Computer Science
Examination paper for: IT3708 Bio-Inspired Artificial Intelligence
Examination date: 15.05.2023
Examination time (from-to): 15:00 – 16:30

Permitted examination support material: D / No printed or hand-written support material is allowed. A specific basic calculator is allowed.

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OTHER INFORMATION

Make your own assumptions: If a question is unclear/vague, make your own assumptions and specify them in your answer. Only contact academic contact in case of errors or insufficiencies in the question set.

Cheating/Plagiarism: The exam is an individual, independent work. During the exam it is not permitted to communicate with others about the exam questions, or distribute drafts for solutions. Such communication is regarded as cheating. All submitted answers will be subject to plagiarism control.

Citations: There is no requirement for citations, given the limited examination time. But if you think that it is natural and useful to include one or more citations on a question, feel free to do so.

Notifications: If there is a need to send a message to the candidates during the exam (e.g. if there is an error in the question set), this will be done by sending a notification in Inspera. A dialogue box will appear. You can re-read the notification by clicking the bell icon in the top right-hand corner of the screen.

Weighting: Each "multiple choice" question has exactly one answer and is worth 1 point, for a total of 18 points. The "written answer" questions are worth a total of 12 points. For each "written answer" question, the maximum score is given with the question. The total number of points for the exam is 30.

ABOUT SUBMISSION

Withdrawing from the exam: If you become ill, or wish to submit a blank test/withdraw from the exam for another reason, go to the menu in the top right-hand corner and click "Submit blank". This cannot be undone, even if the test is still open.

Accessing your answer post-submission: You will find your answer in Archive when the examination time has expired.

1 Multi-objective evolutionary algorithms (MOEAs) make up an important part of bio-inspired AI. Among the alternatives below, pick the wrong or false option.

Select one alternative:

- A goal of some MOEAs is to maximize the diversity of the individuals that we find in an approximated Pareto set.
- A goal of some MOEAs is to maximize the number of individuals found within a certain distance from the true Pareto set.
- A goal of most MOEAs is to maximize the efficiency of optimizing multiple objectives added together in a linear combination.
- A goal of some MOEAs is to minimize the average distance between the approximated Pareto set, as computed by the MOEA, and the true Pareto set.
- A goal of most MOEAs is to jointly optimize several potentially contradictory objectives.

Maximum marks: 1

2 Different chromosome representations are used in bio-inspired AI. Consider the following quote from the literature:

“In the first of these problems, the order in which events occur is important. This might happen when the events use limited resources or time, and a typical example of this sort of problem is the production scheduling problem. […] Another type of problem depends on adjacency and is typified by the traveling salesman problem (TSP).”

Which chromosome representation, often used in these types of problems, is described by the quote above? (Pick one type of representation below.)

Select one alternative:

- Representations in the form of arbitrary computer programs
- Real-valued vectors
- Binary representation
- Integer-valued vectors
- Permutation representation ✔

Maximum marks: 1

3 Among the following statements, which one is not giving a good rationale for the use of niching in evolutionary computation?

Select one alternative:

- It is necessary for multi-objective optimization ✔
- Be realistic from a biological perspective
- Improve the diversity of the population
- Reduce the chance of premature convergence
- Try to ensure high entropy in a population

Maximum marks: 1

4 Consider the following quote from the bio-inspired AI literature:

“Now that each individual in the population has a crowding distance, we use it as a secondary sorting parameter for obtaining the rank of each individual. … [We] rank each individual on the basis of its nondomination level, but we also include a more fine-grained ranking level on the basis of crowding distance.”

Which algorithm is described in this quote?

Select one alternative:

- Particle swarm optimization
- Probabilistic crowding
- NSGA-II ✔
- Deterministic crowding
- Simple genetic algorithm

Maximum marks: 1

5 The idea of evolving multiple subpopulations in tandem is known as island model evolutionary algorithms or coarse-grained parallel evolutionary algorithms. Among the following statements, which one is wrong as it does not describe an obvious benefit or characteristic of the island model?

Select one alternative:

- Multiple subpopulations, one per island, can be evolved in parallel, thus increasing computational efficiency
- Global fitness sharing across all islands is computationally efficient, compensating for the fact that parallel evolution of populations, one per island, may be computationally inefficient
- When communication between islands takes place, there is the injection of individuals of potentially high fitness and likely very different genotypes, thus facilitating exploration via recombination
- During epochs with no communication between islands, subpopulations are evolving independently, and beneficial exploitation within each subpopulation takes place
- Different hyperparameters can be used on different islands for the different subpopulations, thus improving the chance of evolving an overall diverse population

Maximum marks: 1

6 In evolutionary algorithms we often find selection methods. Among the following statements about selection, pick the true statement.

Select one alternative:

- Survivor selection needs to be tightly coordinated with crossover
- Parent selection is essential, while survivor selection (or replacement) is often skipped
- Parent selection needs to be tightly coordinated with mutation
- Parent selection typically does not depend on the chromosome representation
- Survivor selection (or replacement) has minimal impact on how an evolutionary algorithm operates

Maximum marks: 1

7 Which of the following statements is wrong (or false) about evolutionary algorithms (EAs)?

Select one alternative:

- To obtain strong EA results, one should carefully balance exploration and exploitation
- One needs to use EA variants that are optimized to the problem at hand
- Vanilla EAs, such as the simple genetic algorithm, never need to be tailored to work well in a particular application
- Encodings that are suitable to the problem at hand (TSP, …) are typically better to use than the general ones (such as bitstrings)
- It is useful to encode constraints, if present, that work with an EA's representation

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Maximum marks: 1

8 Suppose that we use a permutation representation with a genetic algorithm. Which among the following statements is true, as far as crossover is concerned?

Select one alternative:

- Crossover cannot be used with permutation representations
- Uniform crossover is optimal, as it acts uniformly across the parents
- 1-point crossover is useful, as it always creates valid permutations from valid permutations
- Order 1 crossover is useful, as it creates valid permutations from valid permutations
- For permutation representations, crossover can only be used in combination with mutation

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Maximum marks: 1

9 Consider the following statements about NSGA-II and select the one that is false.

Select one alternative:

- ◯ In NSGA-II the offspring population is first created from the parent population, and then the non-dominated front is formed by sampling uniformly at random from the offspring population. ✔
- ◯ Instead of finding the non-dominated front of the offspring population only, the two populations (offspring and parent) are combined, and then non-dominated sorting is used to classify the entire population.
- ◯ NSGA-II fitness assignment is in the form of ranking based on non-domination sorting.
- ◯ NSGA-II uses fast non-dominated search as a basis for fitness assignment; this can be considered a type of elitism.
- ◯ An explicit diversity-preserving mechanism, crowding distance, is used in NSGA-II.

Maximum marks: 1

10 Consider the following quote from Dan Simon's textbook:

"Not only do we have to initialize a population, as with every other evolutionary algorithm, but we also have to initialize the population's velocity vectors. There are several ways to initialize velocities. For example, they could be initialized randomly, or they could be initialized to zero."

What type of algorithm is described in this quote?

Select one alternative:

- ◯ Genetic algorithm for continuous optimization
- ◯ Particle swarm optimization algorithm ✔
- ◯ Ant colony optimization algorithm
- ◯ None of the other alternatives
- ◯ Evolution strategies for continuous optimization

Maximum marks: 1

11 Crossover is an important operator in evolutionary algorithms. Many crossover variants work both for binary and continuous evolutionary algorithms. Which crossover operator among the following can be used only for continuous evolutionary algorithms:

Select one alternative:

- Segmented crossover
- Uniform crossover
- Multi-parent crossover
- Flat crossover ✔
- Single-point crossover

Maximum marks: 1

12 What is the difference between the Simple Genetic Algorithm (SGA) and the Steady-State Genetic Algorithm (SSGA)?

Select one alternative:

- The SSGA always uses roulette wheel selection
- Typically, SSGA replaces only the worst individual(s), while SGA replaces the entire population ✔
- The SGA always uses binary representations while the SSGA is used for continuous domains
- The SGA only uses mutation, while SSGA uses both mutation and crossover
- The SGA does not use crossover

Maximum marks: 1

13 Which of the following statements is not true about the selection mechanism in EAs?

Select one alternative:

- Selection of survivors is usually deterministic, in that the same number of individuals survives for each generation
- Selection of parents is usually probabilistic, so high quality solutions are likely to be selected but it is not guaranteed
- Selection pushes the population towards higher fitness
- The role of selection is to identify individuals to be selected as either parents or survivors
- Selection operators are dependent on how individuals are represented in the EA ✔

Maximum marks: 1

14 Which of the following statements is **not true** about the single point crossover (1PX) when applied in constrained optimization?

Select one alternative:

- 1PX works for both continuous and binary representations in constrained optimization.
- 1PX can never keep together genes from opposite ends of the string.
- By using 1PX, genes that are near each other are more likely to be kept together.
- Using 1PX on two feasible individuals (when dealing with permutation vectors) ensures that the crossover results in at least one feasible solution. ✔
- 1PX is an exploitation technique.

Maximum marks: 1

15 Which of the following statements is not true about crowding?

Select one alternative:

- Crowding can be generalized by using a scaling factor to alter the odds of selecting an individual
- Crowding can be deterministic, always choosing the fittest individual
- Crowding can occur in replacement, by grouping individuals of similar fitness to make them compete against each other
- Crowding can be probabilistic, more likely choosing the fittest individuals
- Crowding is an implicit approach for keeping population diversity in an EA ✔

Maximum marks: 1

16 Which of the following statements is not true about niching and fitness sharing?

Select one alternative:

- Fitness sharing helps preserving diversity by restricting the number of fit individuals in a niche
- By using fitness sharing, individuals who are far from the rest of the population get an advantage in fitness
- Fitness sharing reduces the fitness of isolated individuals, giving a higher likelihood of survival to those individuals in a crowded neighborhood ✔
- Niching is an approach for keeping population diversity in an EA
- Niching via sharing can be used in EAs to search for multiple optima

Maximum marks: 1

17 Evolutionary algorithms (EAs) can be used to optimize or solve different types of problems. Which of the following statements about problems is not true?

Select one alternative:

- EAs were designed for free optimization problems
- EAs cannot solve constrained optimization problems ✔
- A constrained optimization problem considers a search space, an objective function and a set of constraints that need to be satisfied
- A free optimization problem considers a search space, an objective function, but no set of constraints that need to be satisfied
- A constraint satisfaction problem considers a search space, a set of constraints that need to be satisfied, but no objective function

Maximum marks: 1

18 Which of the following statements is not true about Evolutionary Strategies (ES)?

Select one alternative:

- During the recombination step of ES, a single child is created either locally (from two parents) or globally (two or more parents for each position)
- In ES, the chromosome typically consists of two parts: the object variables and the mutation step sizes
- In a (μ, λ)-ES, λ children are created, and the best μ individuals out of the entire population (including the parents) are selected for survival ✔
- Varying the mutation step size using the 1/5 rule is an adaptive type of parameter control
- In ES, different individuals often have different mutation step sizes

Maximum marks: 1

19 Discuss the pros and cons of using a permutation representation rather than a bitstring representation for the traveling salesperson problem (TSP). Make sure to mention 2 advantages of the permutation representation and describe each advantage by means of 3-5 sentences. (4 points maximum.)

Fill in your answer here

Format

Words: 0

Maximum marks: 4

20 Describe briefly, in 2-3 sentences, what a “memetic algorithm” is. Then describe 2 different ways in which an evolutionary algorithm can be made into a memetic algorithm, using 3-5 sentences for each way. (Maximum 4 points.)

Fill in your answer here

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Words: 0

Maximum marks: 4

21 Parameter tuning is an important part of bio-inspired AI, as it reflects in a model's performance. However, it also has some limitations. Describe briefly at least two limitations of parameter tuning within bio-inspired AI, using 3-5 sentences per limitation. (Maximum 4 points.)

Fill in your answer here

Format

Words: 0

Maximum marks: 4